

Design and development Matrix

CNW41 - cai, DOM35 - Dom, LCS78 - liam, JRS222 - Jonny

Needs analysis		Target specifications			Verification		
Ref Need	Description	Ref Spec	Description	Objectives measurable + tolerance	Verification test description	Location/Team	Control methodology
1	Clear and understandable weather symbols	1.1	Symbols must be big enough to decipher - especially for those with bad vision.	Symbols must fill a 100+/-40mm^2 area	When designing on inventor, use the measure tool on the x and y lengths that the symbol fits within and calculate the area.	Dom on inventor	Area is 100 +/- 40 (mm ²) 1 sample - 100% completion rate
		1.2	Symbols must be universally recognised	Symbols should be recognised by at least 90% of people	Show the symbols to others in the lab and record how many of the symbols are recognised.	2 East lab, Dom, Cai	Symbols are recognised 10 samples - 90% completion
2	Strong and durable casing	2.1	Casing should be able to sustain being dropped from typical heights without damage.	Casing should show no visible damage after being dropped 10 times from 1m up	Drop from 1m (from bottom edge of box which is flat) and check whether visible damage such as cracking or snapping are visible by eye.	2 East lab, Dom, Cai	No visible damage 10 samples - 100% completion
3	Bright Display	3.1	The acrylic sheets must be illuminated enough that the symbols are clear but not uncomfortably bright	Each acrylic sheet should emit between 400+/-100 lux.	Measure the brightness of an acrylic sheet when lit up by each LED and record the luxes using a lux meter.	Dark room, Liam and Jonny	readings of 400 +/- 100 (lux) 1 sample (from each LED) - 100% completion
4	Display current weather	4.1	WIFI connection must remain stable so that display matches online weather API	The display must respond to changes in forecast within 1 minute.	Set up script on PC to measure time between online API updating and the arduino registering this change and sending back a confirmation message to the PC.	2 East lab, Cai and Jonny	expected and accurate outputs based off test script 1 sample - 100% completion
5	Safe to use	5.1	Must have a safe operational voltage	Max Operational voltage of 6V	Connect to power and test with multimeter across each resistor and MOSFET across drain and source pins aswell as LEDs (For each LED when corresponding LED is lit)	2 East lab, Liam	No voltage readings above 6V 1 sample per component - 100% completion
6	Minimal impact on environment	6.1	Low power usage when on	Power rating of 2+/-1W	Utilising a USB power meter and logging power drawn over an hour and finding the average power drawn	2 East lab, Dom and Liam	Average power of 2 +/- 1W 5 samples - 100% completion
7	Sustainability	7.1	The device should be easy to disassemble for recycling	Can be disassembled into base components (acrylic sheets, PCB, arduino, MDF) within 5 minutes	Perform a timed teardown of the final prototype. From fully assembled and plugged in (working state)	2 East lab, Jonny	Time of less then 5 minutes 5 samples (separate people) - 80% completion

		7.2	No exposed wiring or high voltage components	No exposed conductive parts accessible by hand	Visual inspection of the casing and internal wiring. Then reassembled and checked	2 East lab, Dom	No Visual components / wiring 3 samples - 100% completion
8	Assembly and Finish	8.1	Acrylic sheets should be easily swappable/removable	Average removal time < 1 minute without tools	Time a user removing and replacing the acrylic sheets with a new set of 6 sheets.	2 East lab, Cai, Dom and Jonny	Less than 1 minute time 5 samples - 80% completion
		8.2	Casing must have a professional finish	Visual Surface roughness and gaps < 0.5mm	Visual inspect surface for any visible bumps/ craters, then using feeler guages check gaps between lid and case	2 East lab, Dom	No visual bumps / craters, feeler guage test shows gaps < 0.5mm 1 sample - 100% completion